Midterm Quiz

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Exam: TV Program Recommendation

Task

Use the SWEET.TV competition data to produce a ranked Top-5 list of TV programs for each test user.

Implementation

- Implement at least two methods: Content-Based Filtering (CBF), Association Rules, Matrix Factorization (CF), or Personalized PageRank.
- Connect viewing logs with the program schedule to construct user–program interactions. Explain your matching rule and treatment of missing program IDs.
- Compare methods using the same temporal validation split, candidate pool, and evaluation protocol.
- Include a popularity baseline as a reference; it does not count as one of the two required methods.
- Your final submission may use a single method or a hybrid strategy. Evaluate each selected method individually before combining them.
- Use the provided program metadata, but do not use future user viewing behavior for training.

Business & Algorithmic Analysis

Which recommendation strategy would you select for SWEET.TV, and why?

- Select two users and justify your selection. Use the same users across all implemented methods.
- Show each method's Top-5 with program IDs, titles, and categories. Compare the recommendations with the users' viewing histories and program attributes.
- Explain how and why your selected methods differ, and what those differences imply for the user experience.
- Justify your final strategy using validation performance, computational cost, and specific recommendation examples.
- Discuss limitations and failure cases supported by your results. No live A/B test or demonstrated revenue impact is required.

Submission Deadline: GSB Groups - 24th Oct 19:00, AVITO Groups - 25th Oct 19:00, FKN Groups - 1st Nov 19:00

Submit only:

1. Your report in PDF format.
2. Your original, unmodified AI-session export.

The PDF report must include:

- A link to your GitHub repository.
- A screenshot showing the highest returned Kaggle private score for your designated final submission.
- Your method comparison, validation results, and business and algorithmic analysis. Same like Homework.

The GitHub repository must contain:

- Your implementation code.
- Instructions for reproducing the results. (Optional)
- The final submission file used to obtain the reported Kaggle score.

Grading

(Gate0 is automatic fail of exam, please check the homework guideline carefully!)

Kaggle leaderboard-score ranking among exam participants — 50 points

Participants are ranked by the returned score of their designated final submission, not by their position on the frozen Kaggle leaderboard.

Successive 20% groups receive 50 / 40 / 30 / 20 / 10 points.

Homework-style evaluation — 50 points

The existing homework evaluation criteria (c1 and c2) apply, covering implementation, the PDF report—including business analysis—and the original, unmodified AI-session export.

MAP@K — Mean Average Precision at K

How well do we place relevant items near the top?

1. Precision at rank r

$$P_u@r = \frac{\sum_{j=1}^{r} rel_u(j)}{r}$$

Look at the first r recommendations.

What proportion are relevant?

2. Average Precision: one user

$$AP_u@K = \frac{\sum_{r=1}^{K} (P_u@r) rel_u(r)}{\min(K, |R_u|)}$$

Scan the list from top to bottom.
At each relevant item, add precision so far.
Normalize by min(K, total relevant items).

Irrelevant items add no points, but lower precision at later relevant ranks.

3. Mean Average Precision: all users

$$\text{MAP@}K = \frac{1}{|U|} \sum_{u \in U} AP_u@K$$

Compute AP@K for each user.

Average those scores for the same method.

$rel_u(r)$: 1 if relevant, 0 otherwise · R_u: ground-truth relevant items · U: evaluated users

Precision: relevance proportion → AP: one user's ranking → MAP: average across users

Convention shown: $\min(K, |R_u|)$ normalization. Match your evaluator's definition.

MAP@5: Correct recommendations, earlier

Two lists can find the same programs — but rank them differently.

One user · 5 recommendations · **Relevant programs: A, B, C**

Method X

A → B → D → C → E

At each correct recommendation, check precision so far:

Rank 1: 1 correct out of 1 → 1.00

Rank 2: 2 correct out of 2 → 1.00

Rank 4: 3 correct out of 4 → 0.75

AP@5 = (1.00 + 1.00 + 0.75) / 3 ≈ 0.917

Method Y

D → A → E → B → C

At each correct recommendation, check precision so far:

Rank 2: 1 correct out of 2 → 0.50

Rank 4: 2 correct out of 4 → 0.50

Rank 5: 3 correct out of 5 → 0.60

AP@5 = (0.50 + 0.50 + 0.60) / 3 ≈ 0.533

Both lists: Precision@5 = 3/5 = 0.60 · Recall@5 = 3/3 = 1.00

AP scores one user: earlier correct items earn a higher score.
MAP averages AP across users — separately for each method.

AP@K convention: sum precision at relevant ranks, then divide by min(K, number of ground-truth relevant items).
Here the denominator is min(5, 3) = 3, not the number of hits. Match your evaluator's convention.

Search

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Late Submission ···



sweet.tv - TV program recommender

Create the algorithm that predicts user's TV program preferences

Click this one!

Overview Data Code Models Discussion Leaderboard Rules Team Submissions

Overview

Start
Nov 18, 2020

Close
Nov 22, 2020

Description

Cinema Challenge Hackathon dedicated to finding solutions for online cinema. The prize pool is $50 000. We invite data scientists, developers, managers, and designers to participate. Participation is possible both individually and as a team, depending on the chosen setting. What awaits the participants? • $10 000 — for first place in every challenge • $5 000 — for second place in every challenge • 500+ participants — useful acquaintances • professional and personal development You are allowed to use any external data source 10 submissions/day Deadline - 21.11.2020, 11.59 PM Your task is to predict the 5 most preferred **tv_show_id**s for each user from submission.csv, from the most preferred (#1) to the least preferred (#5). The preferences were calculated for the users using the subsequent 12 weeks (weeks 31-42) of historical views.

Competition Host
ogurbych

Prizes & Awards
Kudos
Does not award Points or Medals

Participation
69 Entrants
47 Participants
36 Teams
578 Submissions

Tags
MAP@{K}

Table of Contents
Description

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sweet.tv - TV program recommender

Create the algorithm that predicts user's TV program preferences

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Dataset Description

File descriptions

- **dataset11-30week.csv** - the training set. **vsetv_id** is the same as **channel_id** in export_arh_*.csv files.
- **export_arh_10-19.csv** - supplemental information about train data (TV program schedule). Contains tv_show_id - you have to predict 5 of these for the submission.csv user_ids. Note: tv_show_id = 0 stands for irrelevant TV shows, drop them!
- **export_arh_20-30.csv** - supplemental information about train data (TV program schedule). Contains tv_show_id - you have to predict 5 of these for the submission.csv user_ids. Note: tv_show_id = 0 stands for irrelevant TV shows, drop them!
- **export_arh_31-42.csv** - supplemental information about test data (TV program schedule). Contains tv_show_id - you have to predict 5 of these for the submission.csv user_ids. Note: tv_show_id = 0 stands for irrelevant TV shows, drop them!
- **submission.csv** - a sample submission file in the correct format. Substitute 5 zeros with 5 predicted tv_show_ids for each user!

Train data fields

- **user_id** - an anonymous id unique to a given user
- **vsetv_id** - the id of a channel (equal to **channel_id** in export_arch_*.csv)
- **start_time** - start time of channel being watched
- **end_time** - end time of channel being watched

Submit to Competition

File Upload Notebook MCP

sweet.tv - TV program recommender

You have 99 submissions remaining today. This resets in 9 hours.

Uploaded File

Your submission file

submission.csv (59 KiB)

Your submission should be a CSV file with 1260 rows and a header. You can upload a zip/gz/7z archive.

Submission Description

Write your full name in Russian here!!!!
Джин

Your full name (Russian) & Kaggle name

45 / 500

Submit using the Kaggle CLI

```
>_ kaggle competitions submit -c sweettv-tv-program-recommender -f …
```

Cancel Submit

Search

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Create the algorithm that predicts user's TV program preferences



Overview Data Code Models Discussion Leaderboard Rules Team Submissions

Submissions

You selected 0 of 2 submissions to be evaluated for your final leaderboard score. Since you selected less than 2 submissions, Kaggle auto-selected up to 2 submissions from among your public best-scoring unselected submissions for evaluation. The evaluated submission with the best Private Score is used for your final score.

0/2

Submissions evaluated for final score

All | Successful | Selected | Errors

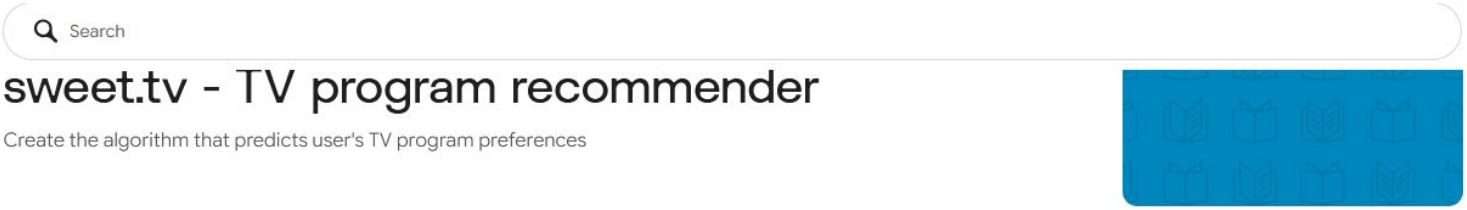
Recent

Submission and Description	Private Score	Public Score	Selected
submission.csv Complete (after deadline) · now · Джин Сеунгмин	0.00000	0.00000	☐
submission.csv Error (after deadline) · 1m ago · Джин Сеунгмин			
submission.csv Complete (after deadline) · 1h ago · template verification	0.00000	0.00000	☐

Error case!!

Please make sure that your name is displayed correctly.

Number of submission is limited to 5 usually. So be wise.



Capture the your Submission page as a screenshot and attach it into the report.

I will accept the best score among them

If you have any issues, please dm to me.

@joseph_jin